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TITLE PAGE

- **Problem Statement ID –** SIH1673
- **Problem Statement Title-** Farmers Disease Diagnostic/Reporting Portal - Mobile Portal AI Based
- **Theme-** Agriculture, FoodTech & Rural Development
- **PS Category-** Software
- **Team ID-** 44478
- **Team Name-** TACTACIANS

IDEA TITLE

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Idea: AI-powered Application for disease diagnosis in Crops and livestocks with a social platform for communal learning

Key-points:

- Enhanced **AI-based** livestock disease diagnosis and necessary solutions.
- **Disease diagnosis** in crops powered by advance **Image based** ML model.
- Powered by a **huge dataset** made with thorough **analysis** and **deep research.**
- Platform for **community engagement** and **Empowerment** of farmers.

TECHNICAL APPROACH

Web App

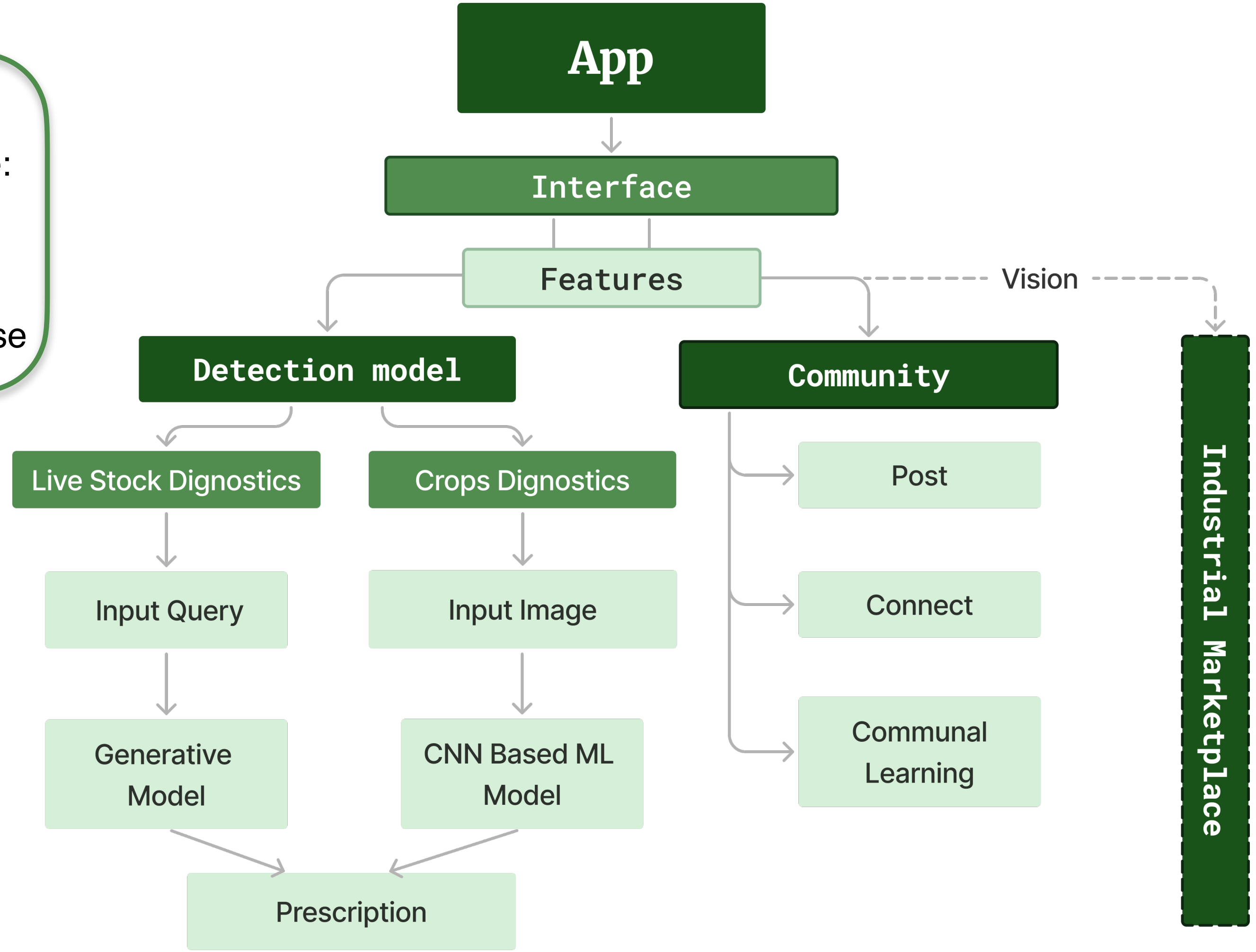
- **Frontend**: HTML,CSS,JS
- **Backend**:NodeJS,ReactJS, ExpressJS
- **Database**: MongoDB

Android App

- **Programming Language**: Java,Kotlin,XML
- **Technologies**: Android Studio,SDK
- **Database**: SQLite,Firebase

Disease diagnosis

- **Programming Language**: Python
- **Technologies**: CNN, TensorFlow/Keras, Streamlit, Pytorch



FEASIBILITY AND VIABILITY

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Environmental/Social Feasibility

Strengths : Empowers farmers; promotes sustainable practices.

Challenges : Trust in AI; cultural and language barriers.

Solutions : Run awareness campaigns; offer localised content in regional languages.

Financial Feasibility

Strengths : Cost effective for farmers; potential government/NGO funding; scalable for broad use.

Challenges : High development and maintenance costs.

Solutions : Public-private partnerships; collaborative business model.

Technical Feasibility

Strengths : Multi-language support, user-friendly interface, Advance AI and image recognition technology, potential NDLM integration.

Challenges : Need of well analysed datasets.

Solutions : Collaboration with institutions for data collection and validation .

IMPACT AND BENEFITS

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Potential Impact on Farmers

- **Faster Diagnosis**: Quick disease detection and targeted treatments.
- **Higher Productivity**: Healthier crops and livestock boost productivity.
- **Cost Efficiency**: Less reliance on experts and better resource use.
- **Knowledge Access**: AI offers expert advice even in remote areas.
- **Empowerment**: Farmers gain more efficiency in decision-making.
- **Loss Prevention**: Early detection reduces disease outbreaks.

RESEARCH AND REFERENCES

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- Neha Chourasia, Surbhi Lanjewar, Shubhangi Daduriya, Mayuri Mahalle, Urvashi Giri , LEAF DISEASE DETECTION USING IMAGE PROCESSING, IJCRT Volume 6, Issue 1 March 2018.

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